

Which dispatch algorithm should Manhattan run, and how sure can we be?

A zone-level digital twin of Uber's Manhattan market, validated against held-out days of real trips, used to compare five dispatch algorithms under bootstrap uncertainty in every input model.

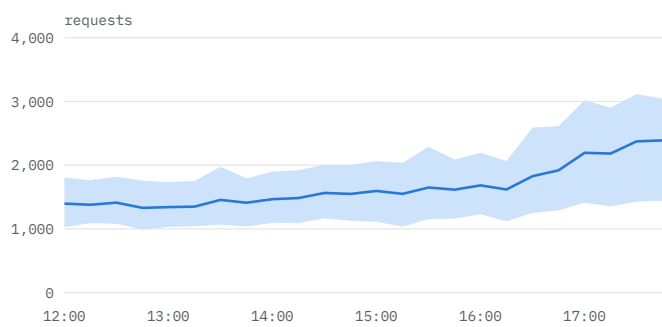
556,297 real trips across 14 weekdays and 69 zones	0.031 held-out wait KS distance, threshold 0.10 PASS	181s median real wait: 117s approach + 45s boarding	39.0% largest wait cut vs greedy: Value-aware + lookahead rebalancing, shortage supply
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1 • THE REAL DATA

Demand climbs toward the evening peak, and so do waits

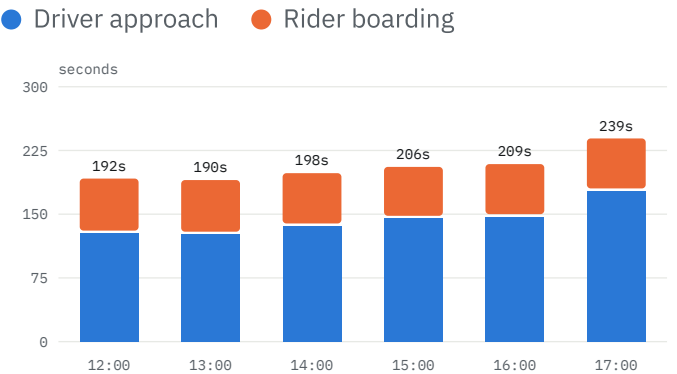
39,736 Uber trips a day start and end in Manhattan between noon and 6pm. The median rider waits 181s and the slowest 10% wait over 358s; at the 17:00 peak at least 3,080 cars are busy with riders at once.

Requests per 15 minutes



Line: mean across the 14 weekdays. Band: the quietest and busiest day in each slot.

Mean wait by hour, split into its two parts



NYC records when the driver arrived on scene, so the wait splits into approach and boarding.

The twin reproduces waits on days it never saw

Input models are fitted on 10 weekdays, the two unobservable supply parameters are calibrated on those same days, and the twin is then scored once against 4 held-out weekdays: KS distance 0.031 on wait times (threshold 0.10) and 0.986 cosine similarity on the demand curve (threshold 0.90). The thresholds were fixed before any real-data run.

Wait quantiles: real held-out days vs simulated



Supply calibration: KS distance by fleet size and same-zone pickup time

45s	0.82	0.74	0.47	0.28	0.24	0.29	0.31	0.34	0.35	0.35
60s	0.83	0.74	0.49	0.21	0.16	0.20	0.23	0.25	0.26	0.26
90s	0.83	0.76	0.51	0.24	0.13	0.07	0.09	0.10	0.11	0.11
120s	0.84	0.77	0.54	0.29	0.19	0.13	0.08	0.06	0.05	0.05
150s	0.85	0.79	0.58	0.37	0.28	0.22	0.19	0.17	0.16	0.16
180s	0.86	0.81	0.62	0.43	0.36	0.31	0.28	0.26	0.25	0.25
240s	0.88	0.83	0.69	0.54	0.49	0.45	0.42	0.40	0.40	0.39
300s	0.89	0.85	0.74	0.63	0.60	0.57	0.53	0.52	0.51	0.51
	1,500	2,500	3,500	5,000	8,000	fleet size (vehicles)				

Darker is a closer fit to the calibration days; outlined cells are within the 0.10 threshold. Every fleet of 4,000 or more fits, so wait data sets a floor on supply but not a ceiling.

How much each algorithm cuts waits, by how many drivers are on the road

Each policy is tuned with the same search budget and must serve at least as many riders as greedy dispatch. Intervals are 95% bootstrap intervals: trips resampled, every input model refitted, all policies run on identical request streams.

shortage · 3,200 cars

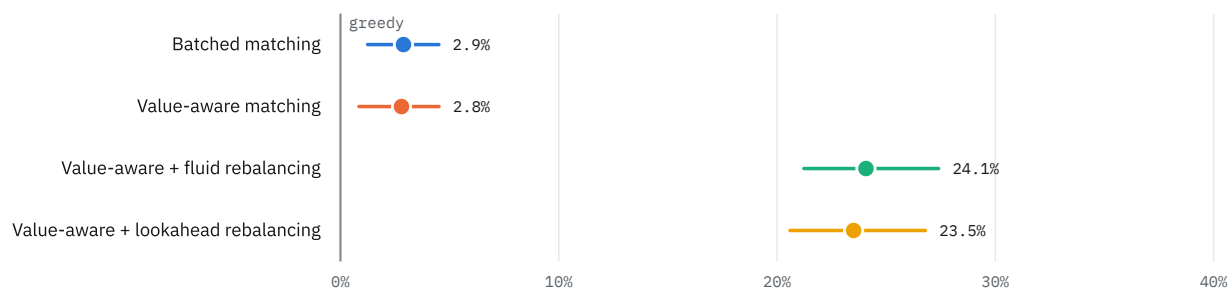
tight · 4,000 cars

mid · 6,500 cars

abundant · 10,000 cars

4,000 cars · 200 bootstrap draws × 5 replications · statistically tied with the best: Value-aware + fluid rebalancing

Wait reduction vs greedy dispatch



POLICY	MEAN WAIT	95% CI	RIDERS SERVED	RANKED FIRST	WORTH IN CARS
● Greedy nearest-vehicle	240s	233s–248s	90.3%	0%	–
● Batched matching	233s	227s–241s	90.4%	0%	144
● Value-aware matching	233s	226s–241s	90.4%	0%	139
● Value-aware + fluid rebalancing (best)	182s	181s–183s	91.9%	100%	≥ 3,787
● Value-aware + lookahead rebalancing	184s	182s–185s	91.8%	0%	≥ 3,787

Wait reduction (%) across every supply regime

POLICY	SHORTAGE · 3,200	TIGHT · 4,000	MID · 6,500	ABUNDANT · 10,000
● Greedy mean wait	314s	240s	231s	222s
● Batched matching	12.0% (9.4–14.6)	2.9% (1.2–4.5)	0.1% (-0.2–0.7)	0.4% (0.1–0.9)
● Value-aware matching	18.0% (14.6–22.0)	2.8% (0.8–4.5)	0.0% (-0.3–0.7)	0.1% (-0.2–0.4)
● Value-aware + fluid rebalancing	38.0% (34.0–42.7)	24.1% (21.2–27.4)	5.1% (4.3–6.4)	1.5% (1.1–2.2)
● Value-aware + lookahead rebalancing	39.0% (35.0–43.7)	23.5% (20.6–26.8)	5.1% (4.2–6.2)	1.6% (1.0–2.2)

Fleet size is the one input the data cannot pin down, so the comparison spans the range it allows, plus a driver shortage below it.

4 · HOW IT WORKS

Method in brief

Simulator. Event-driven, 69 TLC zones, 5-second dispatch ticks. Demand is a non-homogeneous Poisson process per zone and 15-minute bin; destinations, trip times and boarding delays are fitted from the trips.

Algorithms. Greedy nearest-vehicle; batched optimal matching (Hungarian); value-aware matching that prices the drop-off zone with a dynamic-programming value function; and two rebalancing layers that move idle cars toward expected (fluid LP) or sampled (Monte Carlo lookahead) demand.

Uncertainty. Each bootstrap draw resamples the trips, refits every input model and reruns all policies with common random numbers, so differences are paired. A policy is only called

better when its interval excludes zero.

Scale. Exact zone-level reductions (interchangeable vehicles, candidate pruning, a transportation LP for rebalancing) cut a simulated hour from 7.4s to under 1s; the bootstrap ran in parallel on Kaggle.

Limits. Fleet size is identified only from below; riders' patience is assumed (NYC data has no cancellations); zones are treated as points, which misses the very shortest pickups.

Built from results generated 2026-09-29. Data: NYC Taxi & Limousine Commission high-volume FHV trip records.